

Integration Blueprinting

**Blending Business Architecture & Information
Architecture to blueprint an Integration Topology
to support data services**

Overview of the Blueprinting Approach

❑ Step 1: (start with capabilities)

- Start with this worksheet that I put together. The goal is to create a mapping between the business capabilities for CTS and our data objects (core = blue heading AND context = green heading).

❑ Step 2: (map capabilities to data objects)

- Once we can agree on the mapping, the next step is to cross reference with systems and reports to identify the attributes required to support each capability.

❑ Step 3: (identify attributes within each data object)

- We can then conduct a walkthrough with the business (and IT) for each capability and set of data objects to validate the list of attributes. This is where we can delve further into reports, systems, and processes via discussion with SMEs.

❑ Step 4: (identify the sources for attributes)

- Document the SOE (sources of entry for each attribute) and bring the MDM Team in to perform a profile assessment for each attribute within each data object. The assessment should identify the sources of “trust” that we need to establish and what data actually requires mastering.

❑ Step 5: (review / approve mastered core)

- Review the MDM analysis with the business and IT SMEs and get approval from the DDL for the mastered core. That should finalize the game plan for completing the work on the IG.

Mapping Matrix of Key Inputs

Key	Mapping
I	SYSTEM > SYSTEM
I	SYSTEM > REPORTS
M	ATTRIBUTE > OBJECT
M	OBJECT > REPORTS
M	OBJECT > SYSTEM
M	OBJECT > HIERARCHY (structure not an object)
M	OBJECT > MEASURE
M	MEASURE > HIERARCHY
M	OBJECT > REPORTING OBJECT

Index:

I = Interface

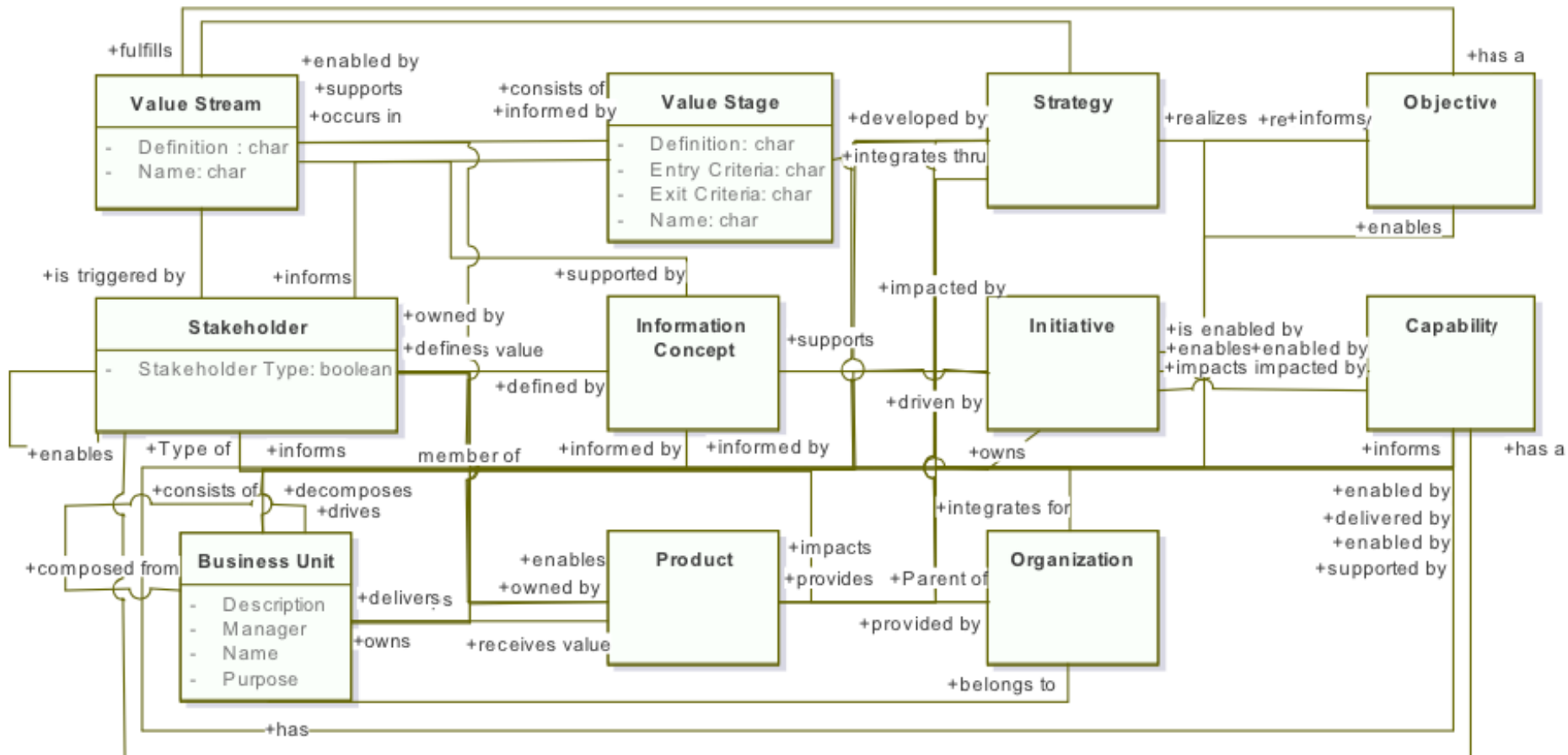
M = Model

Key Transformations: (Mastering)

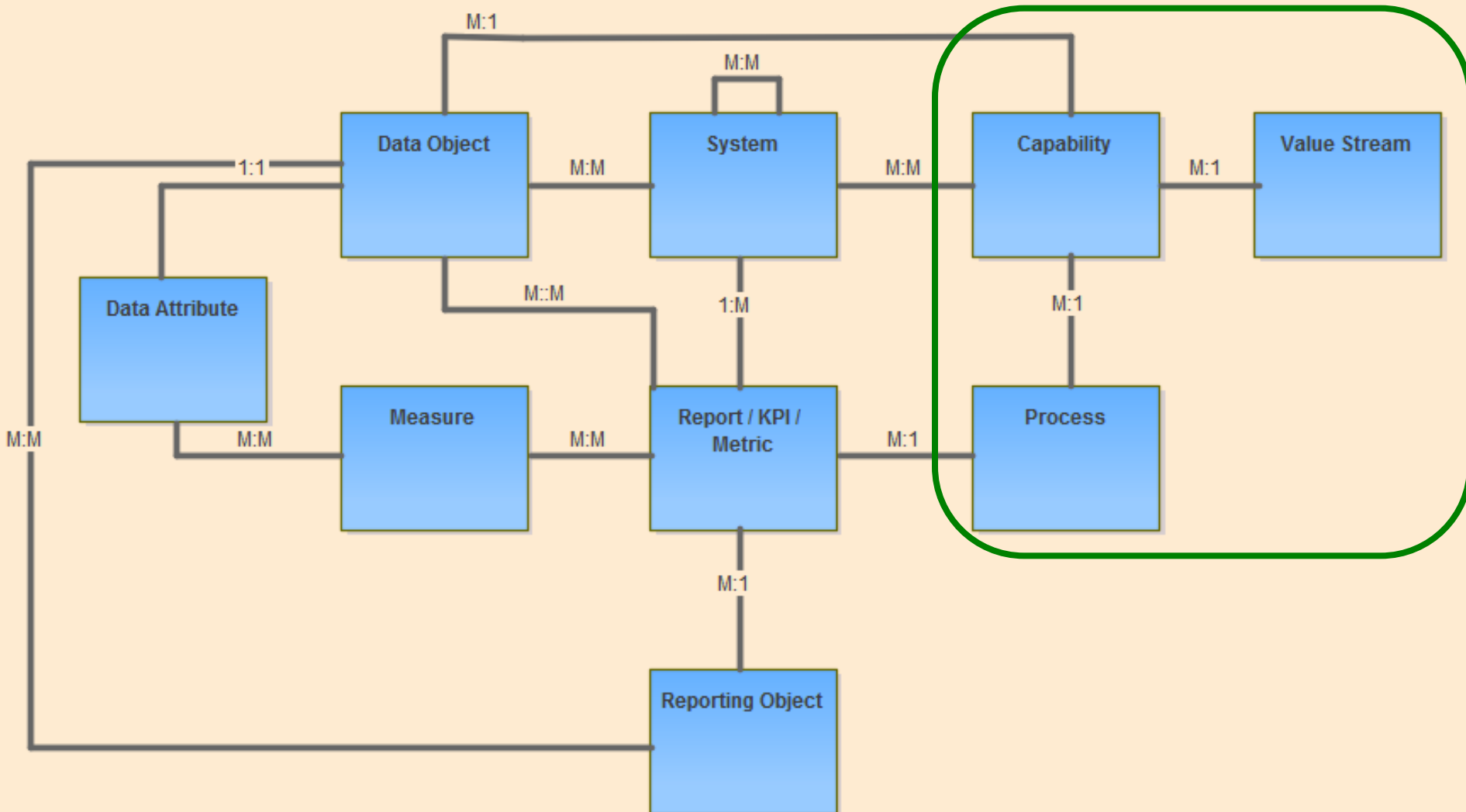
These would be applied to a subset of the mappings listed in the table to the right (highlighted in yellow)

- ☐ Profiling Rules
- ☐ Trust Rules
- ☐ Cleansing Rules
- ☐ Masking Rules
- ☐ Releasability Rules
- ☐ Match / Merge Rules
- ☐ Data Classification Rules
- ☐ Data Inheritance / Cardinality

Step 1: Start with a Business Architecture MetaModel



Step 2: Inputs (MetaModel – BA to Info Object Mapping)



Step 3: Map Capabilities to Information Objects

- ❑ Goals for this step are to ensure traceability and lineage, support impact analysis, ensure there is alignment across teams, and provide separation of ownership (divide and conquer).
- ❑ The diagram on the next slide is a metamodel covering the alignment of business architecture (objects) mapped to information architecture (objects). The business architecture objects are outlined in “green”. The metamodel outlines Key Mappings (and cardinality) between the objects to guide the business analyst through requirements capture and mapping across the business and information architectures.
- ❑ A separate set of tables are provided after the MM slide exploring the mappings and providing a x-ref for architecture teams.

In many cases, we may not have the business architecture for a project, so we can start with the systems information and work backwards from “top” reports, KPIs, and metrics (next best source for list of attributes to be mastered).

Step 3: Map Capabilities to Information Objects

Commercial Technology Strategy [CTS] Capabilities			Data Object Mapping															
Level 1	Level 2	Level 3	Customer	Vendor	Material	Product	Org	Worker	Sales	Territory	Incentive	Inventory	Warehouse	Order	Rebate	B&T	Pricing	Ca
Marketing	Marketing Strategy	Plan/Develop Marketing Strategy	X			X			X	X								
Marketing	Marketing Strategy	Agency Performance/management	X			X			X	X								
Marketing	Market Research and Competitive Intelligence	Market research	X			X	X		X	X								
Marketing	Market Research and Competitive Intelligence	Competitive intel	X				X		X	X								
Marketing	Campaign Strategy and Management	Speaker engagement	X															
Marketing	Campaign Strategy and Management	Campaign Strategy and Management	X			X			X	X								
Marketing	Campaign Strategy and Management	advisory board management	X															
Marketing	Campaign Strategy and Management	Content Management	X															
Marketing	Campaign Strategy and Management	Formulary management	X			X												
Marketing	Campaign Strategy and Management	Launch Project Management	X			X												
Marketing	Campaign Strategy and Management	Customer surveys	X															
Marketing	Campaign Strategy and Management	Marketing Websites	X			X												
Marketing	Campaign Strategy and Management	Launch and Post-Launch Review	X			X												
Marketing	Campaign Strategy and Management	Lead and Opportunity identification	X															
Marketing	Integrated Marketing Planning (Print/Digital Media)	Approve Marketing Material (MARC)	X															
Marketing	Integrated Marketing Planning (Print/Digital Media)	Enable Margin calculation for customers	X															
Marketing	Integrated Marketing Planning (Print/Digital Media)	Marketing Communications	X			X												
Marketing	Packaging Design	Packaging Design	X			X												

- ❑ Capabilities are broken down to Level 3 (in this example) and mapped to the Information Objects (derived from the Business Architecture MetaModel (see prior slide). This ensures that the Data & Information Lifecycle (acquisition through termination) is aligned with the business.

Step 4: Draft a set of Architecture Principles

Architecture Principles		
Principle	Description	Purpose
Primacy of Principles	Architecture principles are applied to all areas of the organization.	Only way to ensure we can maintain a consistent set of measurable decisions related to technology for the enterprise.
Maximize Benefit to the Enterprise	Architecture decisions should be made to benefit the entire enterprise (not just one project or domain).	This attempts to make decisions that result in the greater good rather than benefiting one project or group. Decisions made in this manner will provide more value to all areas of the company, which extends the overall value proposition and helps maximize the ROI. Exception can be made but with the understanding of the negative long term effect. The ARB has a preference for an Enterprise Standard rather than a "One Off".
Traceability	All architecture decisions must trace back to business need	Architecture exists to serve the needs of the business. Therefore, all decisions surrounding the use and purchase of technology should have lineage that is traceable and measurable back to business requirements, which are found in the non-functional features. No everything that IT does is not directly traceable back to the business or a business need, but does provide additional value to the business.
Business Continuity	The business operations should not incur disruption due to technology related decisions	Org. is a global business with 24x7 operations that must be maintained at high levels of uptime and efficiency. This is often dependent upon the application based upon the operational criticality and business risk. Introduction of technology-related changes must be done with careful understanding of risk management guidelines with appropriate mitigation in place to minimize exposure. - Security Governance is derived (in part) from this principle
REUSE	The assets that we build or purchase (outright or via rental) should provide a high degree of common usage (shared) on avg.	This is at the heart of delivering TCO. This is a simple one to discuss, but a very difficult one to manage and enforce. It requires the collective leadership from all levels of the business to embrace this principle to ensure we minimize duplication, inefficiency, and we promote the best services using the fewest number of assets. This applies to data, software and hardware (where the data is owned by the business). Exploit what we have within cost effectiveness rather than buy new.

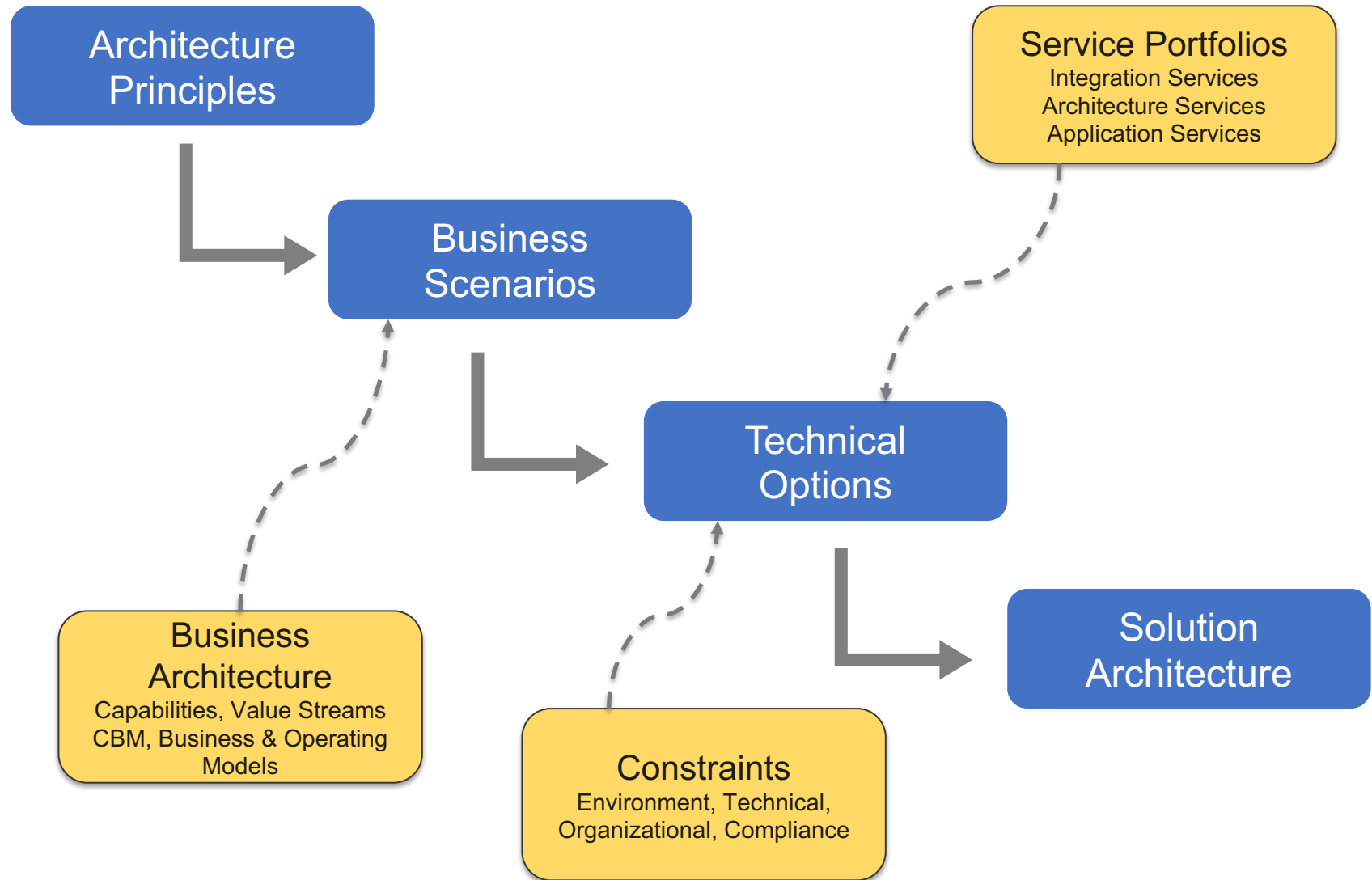
Step 4: Draft a set of Architecture Principles

Architecture Principles		
Principle	Description	Purpose
Compliance	This is related to the strict regulatory constraints and rules which govern our business.	Architectural decisions should assist in meeting regulatory compliance requirements and enhance standards, efficiency and assist in reducing TCO (sustainable and maintainable) by developing tactical implementations of standards. This is very hard to manage and enforce due to the myriad of regulations on a governmental, state and municipal level within multiple countries. IT Risk Management is the POC for all compliance issues. SEE RISK BELOW
Protect Intellectual Property	Intellectual Property must be secured throughout all phases of its lifecycle	While IP is primarily in the business domain, IP is hosted in the IT domain and thus becomes IT's responsibility to ensure the IP (data, content & processes) is governed in a manner that minimizes exposure. Again this needs to be done in a sustainable and maintainable manner. - Data Governance is derived from this principle - Security Governance is derived (in part) from this principle
Abstraction	Application architecture is independent of technology architecture (platform independent) and supports interoperability (integration) independence.	Applications should be built or purchased with portability and abstraction from the physical environment to maximize portability and maintain TCO. In addition, applications should be integrated with other applications and components in a manner that promotes insulation and maximizes interoperability to minimize TCO associated with change management.
Ease of Use	Applications should be easy to use (by business personnel) and abstract users from the underlying technology.	Ease of use lowers the costs of training, transition management and reduces risks associated with attrition and/or organizational realignment. The underlying technology should be transparent to end users. This allows IT to implement technology changes without impacting the end user experience. Do real end user testing before a package is chosen or a series of packages are reused.

Step 4: Draft a set of Architecture Principles

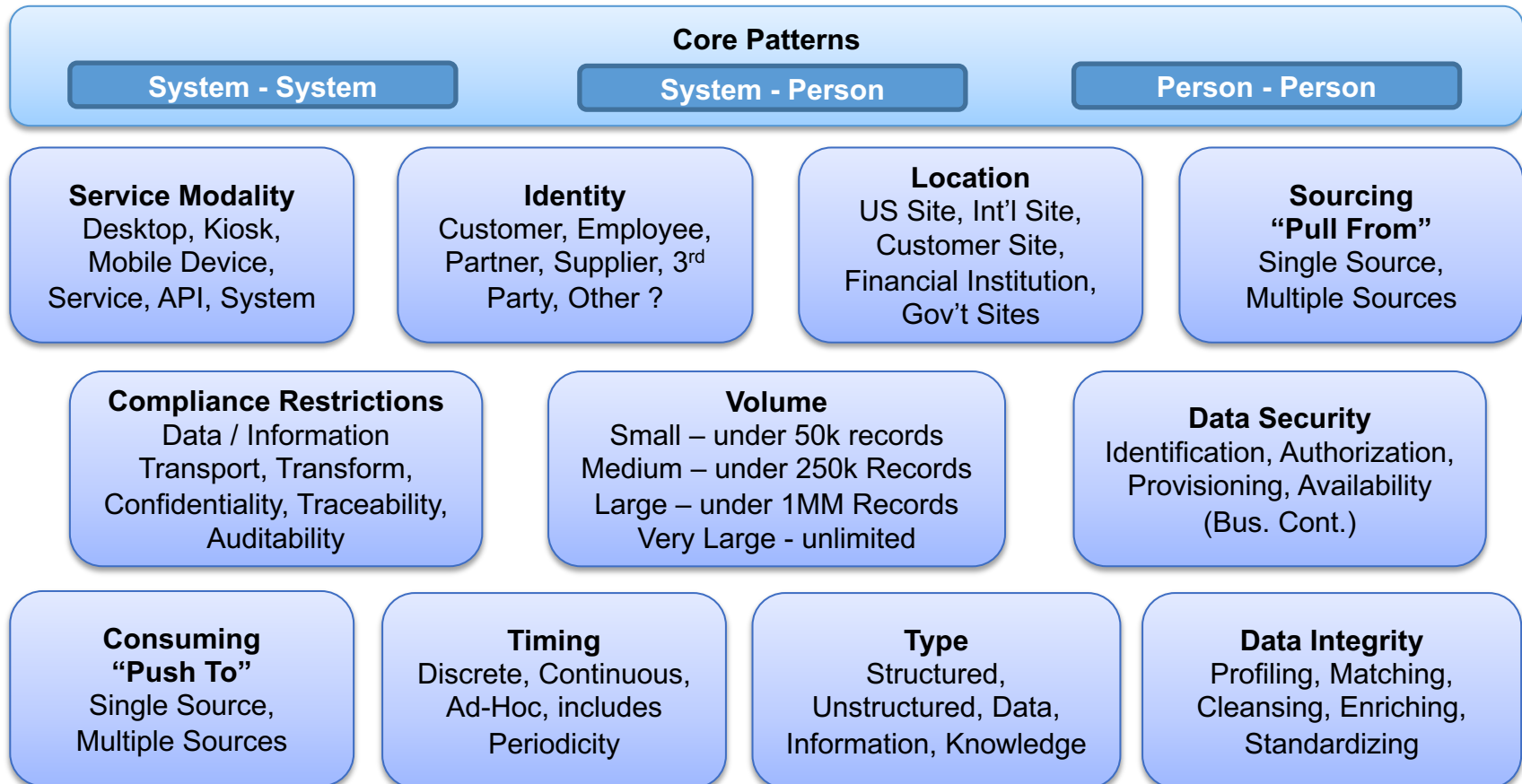
Architecture Principles		
Principle	Description	Purpose
Requirements-Based Change	Technology changes are only made in response to Business Requirements	The business requirements may be related to one or more of the following areas: - Alignment with Corporate Strategy - Increase for Revenues (New or Enhanced Functionality) - Reduced Costs (efficiency or productivity improvements) - Compliance (regulatory requirements and/or Risk Mitigation) - Operational Impact (Process Improvement)
Singularity	A primary goal in obtaining all H/W and S/W is that Hospira standardize on one vendor, application or version	This is about reducing TCO and Support and Personnel costs through the judicious selection of Systems and Software that meet business requirements but have broad applicability for a number of requirements across different business units. Exceptions may be needed for tactical business advantage.
Agility	Design Hospira's architecture so that IT can respond swiftly to changing business demand	Utilizing the principles of abstraction and reuse we should design systems and applications, and manage data an metadata is such a manner to allow IT to respond quickly to meet business requirement are responding to changing market direction or new product innovation with the result being added value by IT

Step 5: Map Inputs that Shape Integration Services



Step 6: Identify the Core Integration Patterns

There are some core patterns which embody the list of integration services, but these are influenced by a number of key considerations, which we can use to derive a list of business scenarios. The list of Key Considerations include (but are not limited to):



Step 7: Identify Technical Options

Any one technical option should be accompanied by a service component in the early stages since the organization and the technology are both maturing and its good practice to provide upfront guidance with any technical option to optimize the solution architecture.

Technical Options

- MDM Only (mods to rules and/or processes)
- MDM + Services (no orchestration)
- MDM + Services (with orchestration)
- Service (no orchestration) [e.g. Simple File Transfer]
- Manual Workaround (stand alone – not supported)



Solution
Architecture



Architecture Services

- Business Architecture Advisory
- Information Architecture Advisory
- Application Architecture Advisory
- Technical Architecture Advisory
- Note: All of these should be considered and more than one may be needed for a solution.

Step 8: Outline a Service Catalogue

☐ Information Lifecycle Mgmt. Services

- Acquisition through Destruction
- Structured & Unstructured
- End-2-End Monitoring (across lifecycle)

☐ Master Data Management Services

- Glossary of Terms
- Semantic & Context Management
- Subject Area Management (Conceptual & Logical)

☐ Dimensional Management Services

- Subject Area Management (Physical)
- Hierarchy rules & governance
- Hierarchy version mgmt. & promotion

☐ Data Integration Services

- Asynchronous access
- Synchronous access
- Batch & Real-Time
- Record, SM Files, VLG Files
- SFTP, Encryption/Decryption
- Application Support
- Web / Mobile Support

☐ Data Integrity Services

- ☐ Profiling, Monitoring, Auditing
- ☐ Cleansing (reactive / proactive)
- ☐ Issue Tracking & Remediation Mgmt.
- ☐ Service Level Alert Mgmt.

☐ Data Security Services

- ☐ Provisioning, Access Control
- ☐ Releasability & Masking
- ☐ Issue Tracking & Remediation Mgmt.
- ☐ Integration with Ent. Threat Mgmt

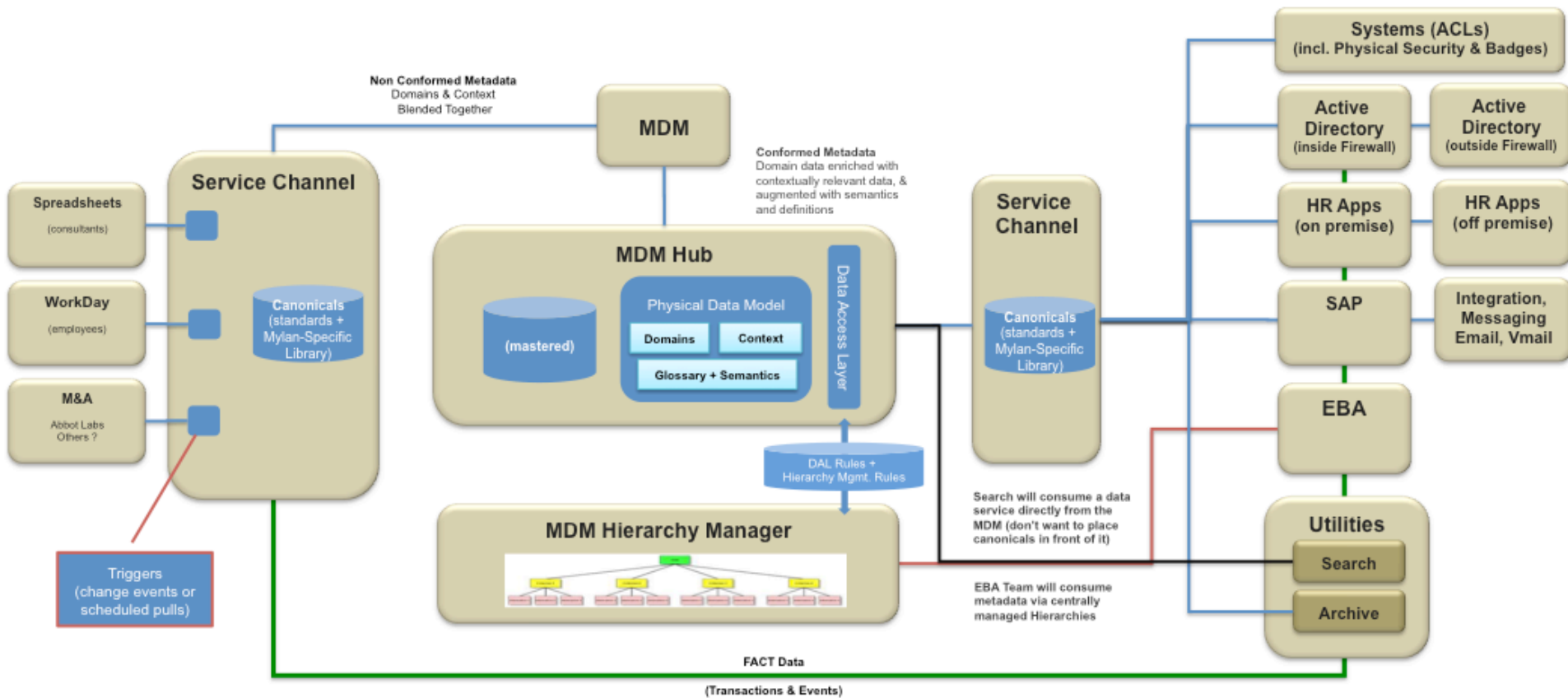
☐ Data / Information Governance Services

- ☐ Policy Development & Mgmt.
- ☐ Policy enforcement (rules)
- ☐ Policy monitoring (metrics & dashboards)

☐ Enterprise Search Services

- ☐ Token-based & Boolean search
- ☐ Metadata Driven Search
- ☐ Semantically enriched matching
- ☐ Natural Language (parsing / matching)
- ☐ Guided search (contextual relevance)

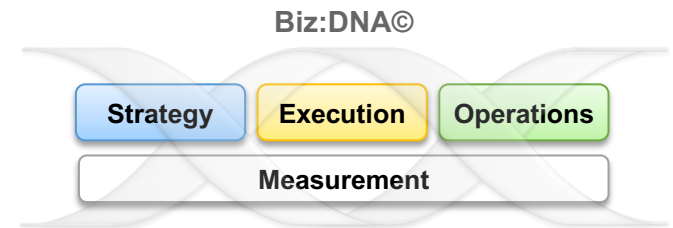
Step 9: Draft an Integration Architecture Blueprint



Example: End-2-End Flow

Thank You

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Build and nurture a winning culture
Proactively manage change & complexity



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